



Protocol for the Examination of Biopsy Specimens from Patients with Gastrointestinal Stromal Tumor (GIST)

Version: 4.4.0.0

Protocol Posting Date: September 2026

The use of this protocol is recommended for clinical care purposes but is not required for accreditation purposes.

This protocol may be used for the following procedures AND tumor types:

Procedure	Description
Biopsy	
Tumor Type	Description
Gastrointestinal stromal tumor	

The following should NOT be reported using this protocol:

Procedure
Resection
Cytologic specimens
Pediatric GIST, familial GIST (germline mutant <i>KIT</i> or <i>PDGFRA</i>) or syndromic GIST

Version Contributors

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Glossary:

Author: Expert who is a current member of the Cancer Committee, or an expert designated by the chair of the Cancer Committee.

Expert Contributors: Includes members of other CAP committees or external subject matter experts who contribute to the current version of the protocol.

Accreditation Requirements

The use of this case summary is recommended for clinical care purposes but is not required for accreditation purposes. The core and conditional data elements are routinely reported. Non-core data elements are indicated with a plus sign (+) to allow for reporting information that may be of clinical value.

Summary of Changes

v 4.4.0.0

- GIST title / lineage is now inclusive of all GIST sites and not limited to stomach
- WHO 6th edition updates to content and explanatory notes
- Reformatted Tumor Site question
- Updates to Histologic Type question, Mitotic Rate instructional Display Item, Treatment Effect Question, and Risk Assessment question

Reporting Template

Protocol Posting Date: September 2026

Select a single response unless otherwise indicated.

CASE SUMMARY: (GASTROINTESTINAL STROMAL TUMOR (GIST): Biopsy)

Standard(s):

This case summary is recommended for reporting biopsy specimens, but is not required for accreditation purposes.

CLINICAL

+Associated Syndrome

- ☐ Carney triad
- ☐ Carney-Stratakis syndrome
- ☐ Neurofibromatosis type1
- ☐ Familial GIST syndrome
- ☐ Other (specify): _____
- ☐ Not specified

SPECIMEN

Procedure

- ☐ Core needle biopsy
- ☐ Endoscopic biopsy
- ☐ Fine needle aspiration biopsy
- ☐ Other (specify): _____
- ☐ Not specified

TUMOR

Tumor Site (Note [A](#))

- ☐ Esophagus (specify location): _____
- ☐ Gastroesophageal junction: _____
- ☐ Stomach (specify location): _____
- ☐ Duodenum
- ☐ Jejunum
- ☐ Ileum (excluding ileocecal valve)
- ☐ Meckel diverticulum (site of neoplasm)
- ☐ Small intestine, NOS
- ☐ Appendix: _____
- ☐ Ileocecal valve: _____
- ☐ Cecum
- ☐ Ascending colon
- ☐ Hepatic flexure of colon
- ☐ Transverse colon
- ☐ Splenic flexure of colon
- ☐ Descending colon
- ☐ Sigmoid colon

- ___ Rectosigmoid junction: _____
- ___ Rectum: _____
- ___ Large intestine, NOS
- ___ Retroperitoneum: _____
- ___ Peritoneum / abdomen (specify site): _____
- ___ Other (specify): _____
- ___ Cannot be determined (explain): _____
- ___ Not specified

Histologic Type

- ___ Gastrointestinal stromal tumor, spindle cell type
- ___ Gastrointestinal stromal tumor, epithelioid type
- ___ Gastrointestinal stromal tumor, mixed
- ___ Gastrointestinal stromal tumor, SDH-deficient
- ___ Gastrointestinal stromal tumor, other (specify): _____

+Histologic Type Comment: _____

Tumor Size (based on clinicoradiologic estimate)

- ___ Greatest dimension in Centimeters (cm): _____ cm
- ___ Cannot be determined (explain): _____

Mitotic Rate (Note B)

The mitotic count should be initiated and reported from the area / lesion that on screening magnification reveals the highest level of mitotic activity and be performed as consecutive high-power fields (HPF). The mitotic rate should be determined in 5 mm² of tumor. With the use of older model microscopes, 50 HPF is equivalent to 5 mm². Most modern microscopes with wider fields require approximately 20 to 25 HPF to encompass 5 mm². If necessary, please measure a field of view to accurately determine actual number of fields required to be counted on individual microscopes to encompass 5 mm².

- ___ Specify mitotic rate per 5 mm²: _____ mitoses per 5 mm²
- ___ Other (specify): _____
- ___ Cannot be determined (explain): _____

Histologic Grade (Note B)

- ___ G1, low-grade (mitotic rate less than or equal to 5 per 5 mm²)
- ___ G2, high-grade (mitotic rate greater than 5 per 5 mm²)
- ___ Other (specify): _____
- ___ GX, cannot be assessed (explain): _____

+Necrosis

- ___ Not identified
- ___ Present

+Extent of Necrosis

- ___ Specify percentage: _____ %
- ___ Other (specify): _____
- ___ Cannot be determined: _____
- ___ Cannot be determined: _____

Treatment Effect (Note [C](#))

- ☐ No known prebiopsy therapy
☐ Not identified
☐ Not specified
☐ Present

+Percentage of Viable Tumor

- ☐ Specify percentage: _____ %
☐ Other (specify): _____
☐ Cannot be determined: _____

+Percentage of Necrosis

- ☐ Specify percentage: _____ %
☐ Other (specify): _____
☐ Cannot be determined: _____
☐ Cannot be determined (explain): _____

Risk Assessment (Note [D](#))

- ☐ None
☐ Very low risk
☐ Low risk
☐ Moderate risk
☐ High risk
☐ Overtly malignant / metastatic
☐ Cannot be determined (explain): _____
☐ Not applicable

+Tumor Comment: _____

ADDITIONAL FINDINGS

+Additional Findings: _____

SPECIAL STUDIES (Note [E](#))

The CAP GIST Biomarker Template can be used for reporting biomarkers.

+Immunohistochemical Studies (select all that apply)

- ☐ Not performed
☐ KIT (CD117)

KIT (CD117)

- ☐ Positive
☐ Negative
☐ Pending

- ☐ DOG1 (ANO1)

DOG1 (ANO1)

- ☐ Positive
☐ Negative

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___ Pending

___ SDHA

SDHA

___ Intact

___ Deficient

___ Pending

___ SDHB

SDHB

___ Intact

___ Deficient

___ Pending

___ BRAF

BRAF

___ Positive

___ Negative

___ Pending

___ Other (specify): _____

+Molecular Genetic Studies (e.g., KIT, PDGFRA, SDHA / B / C / D, NF1, RAS, or BRAF mutational analysis, or NTRK, BRAF, FGFR1 or other fusion gene analysis)

___ Performed, see biomarker report: _____

___ Performed (specify method(s) and result(s)): _____

___ Pending

___ Not performed

COMMENTS

Comment(s): _____

Explanatory Notes

A. Location

Gastrointestinal stromal tumors may occur anywhere along the entire length of the tubal gut, as well as in extravisceral locations, which include the omentum, mesentery, pelvis, and retroperitoneum.[1,2,3,4,5,6](#) Typically, these tumors arise from the wall of the gut and extend inward toward the mucosa, outward toward the serosa, or in both directions. Lesions that involve the wall of the gastrointestinal (GI) tract frequently cause ulceration of the overlying mucosa. Infrequently, lesions invade through the muscularis mucosa to involve the mucosae. Mucosal invasion is an adverse prognostic factor in numerous studies. Because the anatomic location along the GI tract affects prognosis, with location in the stomach having a more favorable prognosis, it is very important to specify anatomic location as precisely as possible.[5,6](#)

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B. Histologic Grade

Histologic grading in GIST, unlike in soft tissue sarcoma, only takes mitotic rate into account. GIST is generally less proliferative than many other soft tissue tumors and the threshold for separating low from high-grade tumors occurs at 5 mitotic figures per 5 mm².[1,2,3](#)

GX: Grade cannot be assessed

G1: Low grade; mitotic rate $\leq 5/5$ mm²

G2: High grade; mitotic rate $> 5/5$ mm²

The mitotic count should be initiated in an area that on screening magnification reveals the highest level of mitotic activity and be performed as consecutive high-power fields (HPF). Stringent criteria should be applied when counting mitotic figures; pyknotic or apoptotic nuclei should not be regarded as mitosis.

Note: Mitoses should be counted in 5 mm² of tumor.^{2,3} With the use of older model microscopes, 50 HPF is equivalent to 5 mm². Most modern microscopes with wider fields require approximately 20 to 25 HPF to encompass 5 mm². If necessary, please measure a field of view to accurately determine actual number of fields required to be counted on individual microscopes to encompass 5 mm².

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C. Treatment Effect

Gastrointestinal stromal tumors respond well to the newer targeted systemic therapies, imatinib mesylate, and sunitinib malate. The types of treatment effects that have been seen are hypocellularity, myxoid stroma, fibrosis, and necrosis. Nests of viable tumor cells are virtually always observed. Because all of these histologic features can be demonstrated in untreated GIST, it is not possible to know whether these changes are due to treatment or not. As a practical compromise, it is best to report the percentage of viable tumor after treatment.

D. Risk Assessment

Biopsies are suboptimally positioned for GIST risk stratification as these may not include sufficient tumor (i.e., 5 mm²) for mitotic counting and may not sample mitotic “hot spots”. Furthermore, the risk for metastasis or tumor related death presumes that the GIST has been removed. On biopsy, one may attempt to risk stratify a GIST, using location, available material for mitotic count, and clinicoradiologic size into account.

Biopsies are more predictive if overtly high mitotic count/high grade/high risk on biopsy, based on mitoses and clinicoradiologic size yet low mitotic count/low grade on biopsy may underestimate actual mitoses on resection and not be accurate due to sampling. Most GIST is now regarded as having at least some potential for distant metastasis. This concept was originally the result of a National Cancer Institute-sponsored consensus conference that was held in 2002.⁴ More specific data generated by large follow-up studies refined the biologic potential assessment.^{2,3,4,5,6,7} Criteria obtained from those data were adopted in a National Cancer Care Network (NCCN) Task Force report on GIST.⁸ We have adopted the criteria for risk stratification, as indicated in Table 1.^{2,3,4,5,6,7} The scheme includes anatomic site as a factor because small bowel GIST carries a higher risk of progression than gastric GIST of similar size and mitotic activity. This prognostic assessment applies best to *KIT/PDGFR*A mutant GIST, whereas *SDH*-deficient GIST is more unpredictable.⁹ For anatomic sites not listed in this table, such as esophagus, mesentery, and peritoneum, or in the case of “insufficient data,” it is best to use risk criteria for jejunum/ileum.¹⁰

Table 1. Guidelines for Risk Assessment of Primary Gastrointestinal Stromal Tumor (GIST)

Tumor Parameters		Risk of Progressive Disease ^a (%)			
Mitotic Rate	Size	Gastric	Duodenum	Jejunum/Ileum	Rectum

≤5 per 5 mm ²	≤2 cm	None (0%)	None (0%)	None (0%)	None (0%)
	>2 - ≤5 cm	Very low (1.9%)	Low (8.3%)	Low (4.3%)	Low (8.5%)
	>5 - ≤10 cm	Low (3.6%)	(Insufficient data##)	Moderate (24%)	(Insufficient data##)
	>10 cm	Moderate (12%)	High (34%)	High (52%)	High (57%)
>5 per 5 mm ²	≤2 cm	None	(Insufficient data##)	High	High (54%)
	>2 - ≤5 cm	Moderate (16%)	High (50%)	High (73%)	High (52%)
	>5 - ≤10 cm	High (55%)	(Insufficient data##)	High (85%)	(Insufficient data##)
	>10 cm	High (86%)	High (86%)	High (90%)	High (71%)

Adapted with permission from Miettinen and Lasota.⁵ Copyright 2006 by Elsevier.

Defined as metastasis or tumor-related death.

Denotes small number of cases.

Data based on long-term follow-up of 1055 gastric, 629 small intestinal, 144 duodenal, and 111 rectal GIST from the pre-imatinib era.^{2,3,4,6}

Note: See Note B, "Histologic Grade," regarding the number of high-power fields to evaluate.

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E. Ancillary Studies

Immunohistochemical Analysis

Because small-molecule kinase inhibitor therapy is highly effective in the treatment of GIST, it has become imperative to distinguish GIST from histologic mimics, mainly leiomyoma, leiomyosarcoma, schwannoma, and desmoid fibromatosis.^{1,2} Immunohistochemistry is instrumental in the workup of GIST. For the initial workup of GIST, a basic immunohistochemical panel including CD117 (KIT), DOG1 (ANO1), desmin, and S100 protein is recommended. GIST is immunoreactive for KIT (CD117) (95%) and/or DOG1 (>99%).^{3,4,5} KIT immunoreactivity is usually strong and diffuse but can be more focal in unusual cases (Figure 1, A, and B). It is not unusual for GIST to exhibit dot-like perinuclear staining (Figure 1, C), while less commonly, cases exhibit membranous staining (Figure 1, D). These patterns do not correlate with mutation type or response to therapy. Most KIT-negative/DOG1 positive GIST are gastric or extra-visceral and almost invariably harbor a *PDGFRA* mutation.⁶ DOG1 expression is not related to mutational status, and is a useful marker to identify KIT-negative GISTs.^{4,5} Approximately 70% of GISTs are positive for CD34, 30% to 40% for smooth muscle actin, 5% for S100 protein (usually focal), 5% for desmin (usually focal), and 1% to 2% for keratin (weak/focal).⁷ PanTrk immunohistochemistry may be positive in GIST independent of *NTRK* fusion, and this immunostain is not recommended in GIST.

Succinate dehydrogenase (SDH) is a mitochondrial enzyme comprising four subunits (SDHA, SDHB, SDHC, and SDHD) that is involved in the Krebs cycle. Genetic alteration of any of the four subunits results in SDH dysfunction and subsequent loss of SDHB expression by immunohistochemistry. SDH-deficient GISTs are characterized by early onset, gastric location, female predominance, multifocality, a plexiform/multinodular growth pattern, frequent lymph node metastases, and epithelioid morphology; they are *KIT* and *PDGFRA* wild-type.^{8,9} One-half to two-thirds of SDH-deficient GISTs are associated with a pathogenic germline mutation in an *SDHx* subunit gene (*SDHx*), most commonly *SDHA*. A substantial proportion of the remaining tumors demonstrate *SDHC* promoter hypermethylation, particularly in the setting of Carney triad. Rare tumors harbor biallelic somatic *SDHx* alterations or remain molecularly unexplained despite comprehensive testing. Thus, it is recommended that all gastric GISTs be screened for loss of SDH by immunohistochemistry, best accomplished by immunostaining for SDHB, which is lost in all SDH-deficient GIST, independent of the SDH-subunit that is inactivated.^{10,11,12,13} Mutations in *SDHA* are detected in ~30% of SDH-deficient GISTs and loss of expression of SDHA specifically identifies tumors with *SDHA* mutations; other SDH-deficient GISTs show normal (intact) cytoplasmic staining.^{14,15} Patients with SDH-deficient GIST should be referred to a genetic counselor for appropriate workup.

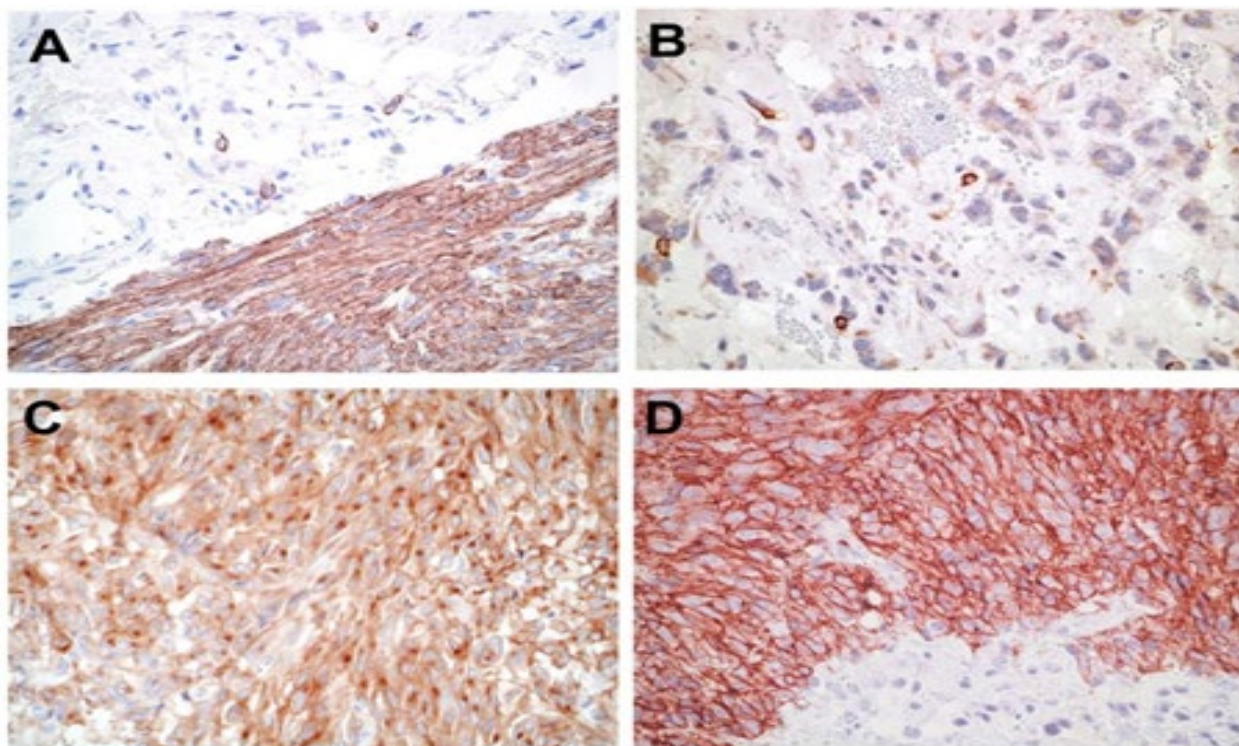


Figure 1. Patterns of KIT staining in gastrointestinal stromal tumor (GIST). A. Diffuse and strong immunoreactivity in a typical GIST. B. Focal and weak staining in an epithelioid gastric GIST with a *PDGFRA* mutation. C. Dot-like perinuclear staining. D. Membranous pattern staining. (Original magnification X400)

Molecular Analysis

Approximately 75% of GISTs possess activating mutations in *KIT*, whereas another 10% have activating mutations in *PDGFRA*.^{16,17,18,19} The resulting mutant proteins demonstrate ligand-independent activation. *KIT* and *PDGFRA* each contain 21 exons. However, mutations cluster within “hotspots”: exons 9, 11, 13, and 17 in *KIT* and exons 12, 14, and 18 in *PDGFRA* (Figure 2). About 10 to 15% of GISTs appear to be negative for both *KIT* and *PDGFRA* mutations. The most recent NCCN Task Force on GIST strongly encouraged that *KIT* and *PDGFRA* mutational analysis be performed in unresectable and metastatic GISTs for which use of tyrosine kinase inhibitors (TKIs) is considered; mutation testing should also be considered for patients with primary disease, particularly if high-risk. *KIT* and *PDGFRA* mutation status can be readily determined from paraffin-embedded tissue. Secondary (acquired) resistance mutations can develop in the setting of long-term imatinib mesylate treatment. These are usually point mutations that occur most commonly in *KIT* exons 13, 14, 17, and 18.²⁰

Recent studies focusing on the molecular classification of GIST recognized two major subgroups: succinate dehydrogenase (SDH)-competent (92-95%) and SDH-deficient GIST (5-8%), both of which can arise in the sporadic or familial setting.^{10,11} SDH-competent GISTs include tumors with mutations of *KIT* or *PDGFRA*, as well as those with mutations in *NF1* (2-6%) or *BRAF* ($\leq 1\%$) and rarely fusions involving *FGFR1* or *BRAF*.^{21,22,23,24,25,26} Other rare molecular genetic events have been described including *PIK3CA* mutation

(which is not necessarily independent of KIT/PDGFRα activation), NTRK fusion (some of these tumors may be better characterized as non-GISTs), and RAS family genes (very uncommon).

SDH-deficient GISTs are characterized by genetic or epigenetic inactivation of any SDH subunit. As mentioned previously, up to two-thirds of SDH-deficient GISTs are associated with a pathogenic germline mutation in an SDH subunit gene (SDHx), most commonly SDHA, with a substantial proportion of the remaining tumors demonstrating SDHC promoter hypermethylation. Somatic biallelic inactivation of an SDHx gene is considered rare.^{27,28,29} The majority of pediatric GIST arise in Carney triad (SDH-deficient GIST, pulmonary chondroma, extra-adrenal paraganglioma; typically due to SDHC promoter methylation) or Carney-Stratakis syndrome (SDH-deficient GIST and paraganglioma due to germline mutation in an SDHx gene). Germline mutations in SDHx can also lead to SDH-deficient renal cell carcinomas and pituitary tumors.

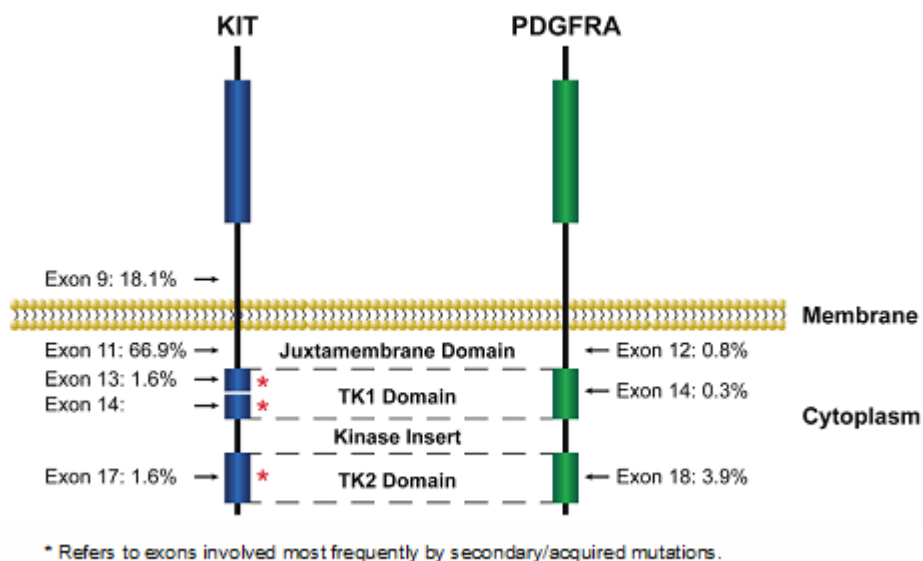


Figure 2. Locations and frequency of activating *KIT* and *PDGFRA* mutations in GIST. Adapted with permission from Heinrich et al.¹⁶ Copyright 2003 by the American Society of Clinical Oncology. All rights reserved.

KIT and *PDGFRA* are excellent targets for small-molecule tyrosine kinase inhibitors (TKIs), and compounds of this class, including imatinib mesylate, sunitinib malate, avapritinib (specifically indicated in *PDGFRA* D842V-mutant tumors), regorafenib, and ripretinib, are Food and Drug Administration-approved for the treatment of GIST.^{30,31,32,33} SDH-deficient GISTs are usually resistant to imatinib but may respond to sunitinib.^{10,34} Clinical interest in mutation status is driven by differential response to TKIs, which may drive drug selection.³⁵ Detection of secondary (acquired) resistance mutations may also impact drug selection as their significance is further defined.

NTRK fusions in GIST are very uncommon. 10 to 15% of tumors may stain positively for pan-TRK, reflecting either wild-type TRK protein expression or non-specific staining, and NGS or FISH are preferred to immunohistochemistry for the detection of *NTRK* fusions in GIST. [36,37,38](#)

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